LCDs provide for crisper images and emit less radiation than CRT monitors, but till recently, they came at an enormous cost. With improvements in manufacturing processes, prices have fallen and you can comfortably include a 15-inch LCD monitor in your PC budget. These now retail for as low as Rs 33,000.

But if you've got your eyes set on a bigger LCD monitor, then every extra inch will cost you a great deal more. A 17-inch LCD will cost you at least Rs 55,000. Move on to the premium segment and you can expect to pay Rs 1 lakh for an 18-inch LCD monitor.

So if crystal clarity is all you want and money is not an issue, then here's how you can buy the best LCD for your money.

Viewable Angle Test

One important consideration when buying an LCD monitor is the angle you can view it at. Most LCD panels display fine when the viewer is seated right in front, but if you are viewing it at an angle, the display quality undergoes a significant amount of change—there is a discernible drop in the brightness level and a yellow or blue tinge appears at more acute angles. So if you plan to use it for presentations or for more than one person to work on it simultaneously, then the viewing angle becomes a critical consideration. The higher the viewing angle supported by the LCD, the better. A presentation on an LCD which supports a viewing angle of less than 120 degrees (combined) will make the presentation look shoddy when viewed from the sides.

In this test Samsung SyncMaster 151 MP topped in the 15-inch category with a score of 9.33 points. In fact, it was the only monitor in this category which allowed us to read the document when sitting at an angle of 80 degrees to the monitor. The drop in brightness was just about tolerable and the document did not become too hazy. Sharp LL-T15V1 and LG Flatron 568 LM stood second with a score of 8 points, whereas Sharp LL-T15S1 performed below average.

In the 17-inch category, Samsung SyncMaster 171MP topped with a score of 14 points and second in line was Convergent with a score of 12.66. These two were the only LCDs that allowed us to read the document somewhat clearly from both the test angles, with no colour variation. Even at an angle of 80 degrees, there was only a slight change in brightness. In

CRT vs LCDs

CRTs are better than LCDs because

- They are less expensive
- They display a wider range of colours
- They have quicker response times which means that they can display moving images without smearing or artefacts
- You can view images from any angle

LCDs are better than CRTs because

- They are lighter and more compact
- They have a lower power consumption
- The image is crisper since each pixel is physically etched on the display
- Eyestrain and fatigue are lower because the images don't flicker
- LCDs generate less low-frequency electromagnetic emissions than CRTs
- The image geometry is always perfect
- The LCD panel reflects less light, which means less glare

Digit Test Process

The LCDs we received were categorised by size as 15-inch, 17-inch, 18-inch and above 18-inches. We tested them at the following test resolutions:

15/17-inch: Recommended resolution (OSD, Auto)

18-inch and above: Recommended resolution (OSD, Auto).

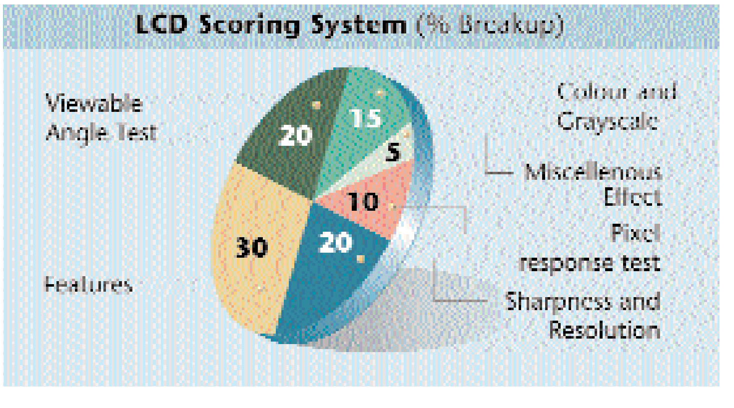
The test bed consisted of a Pentium III 700 MHz processor, 128 MB RAM, and a GeForce2 Ultra 64 MB DDR graphics card. When arriving at the overall score for each monitor, features were given a weightage of 30 per cent and performance, 70 per-cent.

Among features, we looked for

Pixel size: The lower the pixel size, the better.

Viewing angle (horizontal and vertical): LCDs with a viewable angle of 160 degrees and above were given maximum points.

Power Consumption: LCD monitors with a power consumption of 50 watts and less were given maximum points.



The tests consisted of the following:

Viewable angle test: A text document and a tiff image were viewed from two different angles—85 degrees and 60 degrees. We looked for colour and clarity when viewed at these angles.

Pixel response: We played *Quake III* and watched a DVD movie (*The Matrix*) on the LCD and looked out for any blurring of the image.

Point shape and visibility: This is a DisplayMate test where an image is generated with a black background and a fine dot of one pixel size on it.

Here we checked to see if the shape of the dot was correct (square/round). We also looked for pixel bleeding and purity of colour.

Colour and grayscale: This test comprises nine subtests, including streaking and ghosting, bar streaking, mid-range streaking, colour streaking, white level shift, black level shift, and red, green and blue colour purity.